Now we'll introduce a concept called the *wave function*. For now, ignore the name; in general, the quantum wave function may have nothing to do with waves. Later, when we study the quantum mechanics of particles (Lectures 8–10), we'll find out about the connection between wave functions and waves.

Suppose we have a basis of states for some quantum system. Let the orthonormal basis vectors be called $|a,b,c,\ldots\rangle$, where $a,b,c,\ldots$ are the eigenvalues of some complete set of commuting observables **A,B,C**, . …. Now, consider an arbitrary state vector $|\Psi\rangle$. Since the vectors $|a,b,c,\ldots\rangle$ are an orthonormal basis, $|\Psi\rangle$ can be expanded in terms of them:

$$|\Psi\rangle = \sum_{a,b,c,\ldots} \psi(a,b,c,\ldots)|a,b,c,\ldots\rangle.$$

The quantities $\psi(a,b,c,\ldots)$ are the coefficients that enter the expansion. Each of them is also equal to the inner product of $|\Psi\rangle$ with one of the basis vectors:

$$\psi(a,b,c,\ldots) = \langle a,b,c,\ldots|\Psi\rangle. \qquad (5.2)$$

The set of coefficients $\psi(a,b,c,\ldots)$ is called the wave function of the system in the basis defined by the observables **A,B,C**,. … The mathematical definition of a wave function is given by Eq. 5.2, which seems formal and abstract, but the physical meaning of the wave function is profoundly important. According to the basic probability principle of quantum mechanics, the squared magnitude of the wave function is the probability for the commuting observables to have values $a,b,c,\ldots$:

$$P(a,b,c,\ldots) = \psi^*(a,b,c,\ldots)\ \psi(a,b,c,\ldots).$$

The form of the wave function depends on which observables we choose to focus on. That's because calculations for two different observables rely on